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System and method for forwarding network traffic

Abstract

A device for control of network traffic may include a plurality of edge interface circuit and internal interface circuits each coupled to one or more network components. The device may prepend frame identification information to received data frames and remove duplicate data frames when identification information is detected multiple times. The device may store frame identification information in a non-transitory memory device and perform a lookup operation to identify duplicate data frames and eliminate loops in the network.

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Background/Summary

PRIORITY (1) This application claims priority to commonly owned U.S. Patent Application No. 63/404,736 filed Sep. 8, 2022, the entire contents of which are hereby incorporated by reference for all purposes.

FIELD OF THE INVENTION

(1) The present disclosure relates to forwarding of traffic in a computer network.

BACKGROUND

(2) In a computer network, network components, including but not limited to switches and routers, are tasked with receiving data at their input ports, also termed ingress ports, and forwarding that received data through their output ports, also termed egress ports, and on to the proper destination. This process of sending incoming data on to the appropriate set of destinations is termed forwarding in computer networking.

(3) In one type of network, commonly termed a store-and-forward (S&F) network, a network component may receive an incoming frame at its ingress port. The frame of data is received in its entirety before a forwarding decision is made. Once the entire frame of data is received and buffered, the network component will compare the Cyclic Redundancy Check (CRC) data included in the frame to determine the correctness of the received data. If the CRC check indicates the data is correct, the data is forwarded to the appropriate Media Access Control (MAC) address.

(4) In another type of network, commonly termed a cut-through-forwarding (CTF) network, only a small portion of the incoming frame is received and once the forwarding decision is made, the network component begins to send the incoming frame data to the appropriate egress ports. The forwarding decision may comprise a MAC address destination, or may comprise other information. The network component does not wait for the entire frame to be received before beginning to forward the frame to the appropriate destinations.

(5) Cut-through forwarding is advantageous in many applications as it provides for low latency forwarding through a network component, and this low latency can be fixed and independent of frame size.

(6) Cut-through forwarding may not detect and discard frames exposed to bit-errors before forwarding to the next destination in the network. A first network component may begin to forward the start of a frame before the end of frame is received at its ingress port, forwarding the frame at its egress port to the proper destination at a second network component. This second network component may receive the data frame at its ingress port, and similarly begin to forward the start of the frame before the end of the frame is received at its ingress port, forwarding the frame at its egress port to a third network component. In a similar manner, a data frame may be forwarded across many network components in a CTF network. When the end of the frame is received at the first network component, a CRC check may be performed. If an error is detected, that frame has already been forwarded to at least one destination. In a broadcast or multicast network implementation, erroneous data frames may be forwarded to many destinations in the network before errors are detected. Correcting these erroneous data frames increases the complexity of the network and increases latency. Additionally, a bit error in the frame fields used in the forwarding decision may cause data frames to be looped in the network as data frames are continually forwarded to incorrect destinations.

(7) A mechanism is needed to prevent excess network traffic and unnecessary looping of data traffic in networks where cut-through forwarding is used.

SUMMARY

(8) The examples herein enable a device, system and method for low-latency network forwarding eliminating excessive network traffic and unnecessary looping of data.

(9) According to one aspect, the examples herein enable a device including an edge interface circuit to couple to a network component and to couple to a forwarder circuit. The edge interface circuit receives data frames from the network component, prepends a frame-specific information header to the received data frames and forwards the prepended data frames to the forwarder circuit. The edge interface circuit receives data frames from the forwarder circuit, removes duplicate data frames and forwards non-duplicate data frames to the network component. An internal interface circuit couples to an internal network segment and couples to the forwarder circuit, the internal interface circuit receives data frames from the internal network segment and forwards the received data frames to the forwarder circuit. The internal interface circuit receives data frames from the forwarder circuit, removes duplicate data frames and forwards non-duplicate data frames to the internal network segment.

(10) According to one aspect, the examples herein enable a system including a network. The network includes a plurality of network connected devices. The network connected devices include an edge interface circuit coupled to a network component and coupled to a forwarder circuit, the edge interface circuit receives data frames from the network component, prepends a frame-specific information header to the received data frames and forwards the prepended data frames to the forwarder circuit. The edge interface circuit receives data frames from the forwarder circuit, removes duplicate data frames and forwards non-duplicate data frames to the network component. An internal interface circuit coupled to an internal network segment and coupled to the forwarder circuit, wherein the internal interface circuit receives data frames from the internal network segment and forwards the received data frames to the forwarder circuit. The internal interface

circuit receives data frames from the forwarder circuit, removes duplicate data frames and forwards non-duplicate data frames to the internal network segment.

(11) According to one aspect, the examples herein enable a method including the following operations: receiving a data frame at an edge interface circuit from a network component, generating a frame-specific information header based on the received data frame, prepending the frame-specific information header as a tag onto the received data frame to generate a prepended data frame, forwarding the prepended data frame to a forwarder circuit, receiving a data frame at an edge interface circuit from the forwarder circuit and discarding data frames with a detected error, identifying, based upon contents of a non-transitory memory and the frame-specific information header, duplicate received data frames, removing duplicate received data frames, removing the frame-specific information header from the non-duplicate received data frame to create an updated data frame, forwarding the updated data frame to a network destination through an egress port, receiving a data frame at an internal interface circuit from an internal network segment, forwarding the received data frame to the forwarder circuit, receiving a data frame at an internal interface circuit from the forwarder circuit, identifying, based upon contents of a non-transitory memory and the frame-specific information header, duplicate received data frames, removing duplicate received data frames, and forwarding non-duplicate data frames to a network component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The figures illustrate examples of systems for control of network traffic.
- (2) FIG. 1 illustrates one of various examples of a computer network.
- (3) FIG. 2 illustrates one of various examples of an edge circuit.
- (4) FIG. 3 illustrates a method for control of network traffic.

DETAILED DESCRIPTION

- (5) FIG. 1 illustrates one of various examples of a computer network **100**. The example of FIG. 1 includes network components **110**, **130**, and **140**. Network components **110**, **130** and **140** may include, but are not limited to bridges, switches and routers, without limitation. There is no requirement that network components **110**, **130** and **140** be of the same type. Internal network segment **150** may include one or more network components, the one or more network components utilizing a CTF protocol. Edge circuits **115**, **135**, **145** may connect internal network segment **150** to, respectively, network components **110**, **130**, **140**. Cut-through forwarder circuits **151** and **152** of internal network segment **150** may forward data frames between edge circuits **115**, **135**, **145** using a CTF protocol.
- (6) Network component **110** may be coupled to internal network segment **150** through edge circuit **115**. Edge circuit **115** may include a first edge interface circuit **118** to communicate with network component **110** using a store-and-forward protocol. First edge interface circuit **118** may include at least one ingress port **111** and at least one egress port **112**. First edge interface circuit **118** may receive data from network component **110** via ingress port **111**. First edge interface circuit may send data to network component **110** via egress port **112**. Edge circuit **115** may include one or more internal interface circuits **116**, **117** to communicate with cut-through forwarder circuits **151** and **152** of internal network segment **150**.
- (7) The example illustrated in FIG. 1 includes two cut-through forwarder circuits **151** and **152** of internal network segment **150**, but this is not intended to be limiting. Other examples may include more cut-through forwarder circuits than those illustrated in FIG. 1 or may include fewer cut-through forwarder circuits than those illustrated in FIG. 1.
- (8) The specific configuration of network components and specific paths between network components as illustrated in FIG. 1 is for illustrative purposes and is not intended to be limiting.

(9) Edge circuit **115** may receive a frame of data at first edge interface circuit **118** from network component **110** via ingress port **111**. The frame of data received at edge circuit **115** may contain identification information. Edge circuit **115** may include an error detector circuit **172** to detect errors in the received data frame. Error detector circuit **172** may perform a CRC check or other error detection check on the received data frame. Edge circuit **115** may perform a CRC check or other error detection check on the received data frame. Edge circuit **115** may prepend a frame-specific information header to the received data frame to create a prepended data frame. The frame specific information header may be generated by identification circuit **175**. Identification circuit **175** may be implemented in hardware or software, or a combination thereof. Identification circuit **175** may be integrated into first edge interface circuit **118**, or may be separate component. The frame-specific information header may include the source MAC address for the received data frame. The frame-specific information header may include an identifier associated with the received data frame. This identifier may also be termed a frame number. The identifier or frame number may distinguish a first received data frame, received from a particular MAC address, from a second received data frame received from the same MAC address at an earlier or later point in time. The frame-specific information header may be part of a tag prepended to the received data frame to generate the prepended data frame. The tag may be prepended by tag insertion circuit **176**. Tag insertion circuit **176** may be coupled to identification circuit **175**, and may be implemented in hardware or software, or a combination thereof. Tag insertion circuit **176** may be integrated into first edge interface circuit **118**, or may be a separate component. The prepended data frame with the prepended frame-specific information header may be forwarded to other destinations in the network. The prepended data frame with the prepended frame-specific information header may be forwarded to forwarder circuit **152** of internal network segment **150** via forwarder circuit **113**, internal interface circuit **117** and path **162**. The prepended data frame with the prepended frame-specific information header may be forwarded by forwarder circuit **113** to forwarder circuit **151** of internal network segment **150** via forwarder circuit **113**, internal interface circuit **116** and path **161**.

(10) Edge circuit **115** may receive a frame of data at internal interface circuit **116** via path **161**. The frame of data received at edge circuit **115** may contain identification information. Edge circuit **115** may update portions of the received data frame and forward the updated data frame to network component **110** using a S&F protocol via egress port **112**. Edge circuit **115** may include an error detector circuit **172** to detect errors in the received data frame and discard data frames with a detected error. Error detector circuit **172** may perform a CRC check or other error detection check on the received data frame. If an error is detected, the frame may be discarded. Edge circuit **115** may read the portion of the received data frame comprising the frame-specific information header. The frame-specific information header may be present at a predetermined location in the received data frame. Edge circuit **115** may perform a lookup operation in cooperation with a Remove Duplicates Table (RDT) **171** in edge circuit **115** to determine if the frame-specific information header in the received data frame matches the frame-specific information header of a previous data frame that has already been forwarded. The frame-specific information header may be stored in RDT **171** in edge circuit **115**. RDT **171** may be a non-transitory memory. If the frame-specific information header of the received data frame is listed in RDT **171**, the received data frame may be discarded and not forwarded since it has already been forwarded through egress port **112**. The lookup operation and discarding of duplicate data frames may be implemented by a duplicates removal circuit **173**. Duplicates removal circuit **173** may be implemented in hardware or software, or a combination thereof. If the frame-specific information header of the received data frame is not listed in RDT **171**, the data frame may be termed a non-duplicate data frame, and edge circuit **115** may add the frame-specific information header to RDT **171**. After performing the lookup operation, edge circuit **115** may remove the frame-specific information header from the data frame to create an updated data frame. The removal of the frame-specific information header may be implemented by a tag removal circuit **174**. Tag removal circuit **174** may be implemented in hardware or

software, or a combination thereof. The updated data frame may be forwarded to first edge interface circuit **118** via forwarder circuit **113**. The updated data frame may be forwarded to network component **110** via first edge interface circuit **118** at egress port **112**.

(11) Edge circuit **115** may receive a frame of data at internal interface circuit **117** via path **162**. The frame of data received at edge circuit **115** may contain identification information. Edge circuit **115** may update portions of the received data frame and forward the updated data frame to network component **110** using a S&F protocol via egress port **112** in first edge interface circuit **118**. Edge circuit **115** may include an error detector circuit **172** to detect errors in the received data frame and discard data frames with a detected error. Error detector circuit **172** may perform a CRC check or other error detection check on the received data frame. If an error is detected, the frame may be discarded. Edge circuit **115** may read the portion of the received data frame comprising the frame-specific information header. The frame-specific information header may be present at a predetermined location in the received data frame. Edge circuit **115** may perform a lookup operation in cooperation with RDT **171** to determine if the frame-specific information header matches the frame-specific information header of a previous data frame that has already been forwarded. If the frame-specific information header of the received data frame is listed in RDT **171**, the received data frame may be discarded and not forwarded since it has already been forwarded through egress port **112** of first edge interface circuit **118**. The lookup operation and discarding of duplicate data frames may be implemented by duplicates removal circuit **173**. If the frame-specific information header of the received data frame is not listed in RDT **171**, the data frame may be termed a non-duplicate data frame, and edge circuit **115** may add the frame-specific information header to RDT **171**. After performing the lookup operation, edge circuit **115** may remove the frame-specific information header from the data frame to create an updated data frame. The removal of the frame-specific information header may be implemented by tag removal circuit **174**. The updated data frame may be forwarded to first edge interface circuit **118** via forwarder circuit **113**. The updated data frame may be forwarded to network component **110** at egress port **112**.

(12) Network component **140** may be coupled to edge circuit **145**. Edge circuit **145** may include one or more edge interface circuits **148**, **149** to communicate with network component **140** using a store-and-forward protocol. First edge interface circuit **148** may communicate with network component **140** via ingress port **141** and egress port **142**. Second edge interface circuit **149** may communicate with network component **140** via ingress port **143** and egress port **144**. Edge circuit **145** may include one or more internal interface circuits **146**, **147** to communicate with other network components in internal network segment **150**.

(13) Edge circuit **145** may receive a frame of data at first edge interface circuit **148** from network component **140** via ingress port **141**. Edge circuit **145** may include an error detector circuit **182** to detect errors in the received data frame. Error detector circuit **182** may perform a CRC check or other error detection check on the received data frame. The frame of data received at edge circuit **145** may contain identification information. Edge circuit **145** may prepend frame-specific information header in the received data frame to create a prepended data frame. The frame-specific information header may be generated by an identification circuit **185** of edge circuit **145**. Identification circuit **185** may be implemented in hardware or software, or a combination thereof. Identification circuit **185** may be integrated into first edge interface circuit **148**, or may be a separate component. The frame-specific information header may include an identifier associated with the received data frame. The frame-specific information header may be part of a tag prepended to the received data frame to generate a prepended data frame. The tag may be prepended by tag insertion circuit **186** of edge circuit **145**. Tag insertion circuit **186** may be coupled to identification circuit **185**, and may be implemented in hardware or software, or a combination thereof. The prepended data frame with the prepended frame-specific information header may be forwarded to cut-through forwarder circuit **151** of internal network segment **150** via forwarder

circuit **174**, internal interface circuit **146** and path **166**. The prepended data frame with the prepended frame-specific information header may be forwarded to edge circuit **135** via forwarder circuit **174**, internal interface circuit **147** and path **164**.

(14) Edge circuit **145** may receive a frame of data at second edge interface circuit **149** from network component **140** via ingress port **143**. Edge circuit **145** may include an error detector circuit **182** to detect errors in the received data frame. Error detector circuit **182** may perform a CRC check or other error detection check on the received data frame. The frame of data received at edge circuit **145** may contain identification information. Edge circuit **145** may prepend a frame-specific information header in the received data frame to create a prepended data frame. The frame-specific information header may be generated by identification circuit **185**. Identification circuit **175** may be integrated into second edge interface circuit **149**, or may be a separate component. The frame-specific information header may include an identifier associated with the received data frame. The identifier may also be termed a frame number. The identifier or frame number may distinguish a first received data frame, received from a particular MAC address, from a second received data frame received from the same MAC address at an earlier or later point in time. The frame-specific information header may be part of a tag prepended to the frame of data. The tag may be inserted by tag insertion circuit **186**. Tag insertion circuit **186** may be integrated into second edge interface circuit **149**, or may be a separate component. The prepended data frame with the prepended frame-specific information header may be forwarded to other destinations in the network. The prepended data frame with the prepended frame-specific information header may be forwarded to edge circuit **135** via forwarder circuit **174**, internal interface circuit **147** and path **164**. The prepended data frame with the prepended frame-specific information header may be forwarded to cut-through forwarder circuit **151** of internal network segment **150** via forwarder circuit **174**, internal interface circuit **146** and path **166**.

(15) Edge circuit **145** may receive a frame of data at first internal interface circuit **146** via path **166**. Edge circuit **145** may include an error detector circuit **182** to detect errors in the received data frame and discard data frames with a detected error. Error detector circuit **182** may perform a CRC check or other error detection check on the received data frame. If an error is detected, the frame may be discarded. The frame of data received at edge circuit **145** may contain identification information. Edge circuit **145** may update portions of the received data frame and forward the updated data frame to network component **140** using a S&F protocol via egress port **142**. Edge circuit **145** may read the portion of the received data frame comprising the frame-specific information header. Edge circuit **145** may perform a lookup operation in cooperation with RDT **181** in edge circuit **145** to determine if the frame-specific information header matches the frame-specific information header of a previous data frame that has already been forwarded. The frame-specific information header may be stored in RDT **181** in edge circuit **145**. RDT **181** may be a non-transitory memory. If the frame-specific information header of the received data frame is listed in RDT **181**, the received data frame may be discarded and not forwarded since it has already been forwarded via egress port **142**. Edge circuit **145** may include an error detector circuit **182** to detect errors in the received data frame and discard data frames with a detected error. The lookup operation and discarding of duplicate data frames may be implemented by duplicates removal circuit **183** in edge circuit **145**. Duplicates removal circuit **183** may be implemented in hardware or software, or a combination thereof. Duplicates removal circuit **183** may include a first circuit to detect duplicate data frames forwarded to egress port **142** and a second circuit to detect duplicate data frames forwarded to egress port **144**. If the frame-specific information header of the received data frame is not listed in RDT **181**, the data frame may be termed a non-duplicate data frame, and edge circuit **145** may add the frame-specific information header to RDT **181**. After performing the lookup operation, edge circuit **145** may remove the frame-specific information header from the data frame to create an updated data frame. The removal of the frame-specific information header may be implemented by tag removal circuit **184** in edge circuit **145**. The tag removal circuit **184** may be

implemented in hardware or software, or a combination thereof. The updated data frame may be forwarded to first edge interface circuit **148** via forwarder circuit **174**. The updated data frame may be forwarded to network component **140** via first edge interface circuit **148** at egress port **142**.

(16) Edge circuit **145** may receive a frame of data at internal interface circuit **147** via path **164**. Edge circuit **145** may include an error detector circuit **182** to detect errors in the received data frame and discard data frames with a detected error. Error detector circuit **182** may perform a CRC check or other error detection check on the received data frame. If an error is detected, the frame may be discarded. The frame of data received at edge circuit **145** may contain identification information. Edge circuit **145** may update portions of the received data frame and forward the updated data frame to network component **140** using a S&F protocol via egress port **144**. Edge circuit **145** may read the portion of the received data frame comprising the frame-specific information header. The frame-specific information header may be present at a predetermined location in the received data frame. Edge circuit **145** may perform a lookup operation in cooperation with RDT **181** to determine if the frame-specific information header matches the frame-specific information header of a previous data frame that has already been forwarded. The frame-specific information header may be stored in RDT **181** in edge circuit **145**. RDT **181** may be a non-transitory memory. For data frames forwarded to egress port **144**, if the frame-specific information header of the received data frame is listed in RDT **181**, the received data frame may be discarded and not forwarded since it has already been forwarded through egress port **144**. For data frames forwarded to egress port **142**, if the frame-specific information header of the received data frame is listed in RDT **181**, the received data frame may be discarded and not forwarded since it has already been forwarded through egress port **142**. The lookup operation and discarding of duplicate data frames may be implemented by duplicates removal circuit **183**. For data frames forwarded to egress port **144**, if the frame-specific information header of the received data frame is not listed in RDT **181**, the data frame may be termed a non-duplicate data frame, and edge circuit **145** may add the frame-specific information header to RDT **181**. For data frames forwarded to egress port **142**, if the frame-specific information header of the received data frame is not listed in RDT **181**, the data frame may be termed a non-duplicate data frame, and edge circuit **145** may add the frame-specific information header to RDT **181**. After performing the lookup operation, edge circuit **145** may remove the frame-specific information header from the data frame to create an updated data frame. The removal of the frame-specific information header may be implemented by tag removal circuit **184** in edge circuit **145**. The updated data frame may be forwarded to second edge interface circuit **149** via forwarder circuit **174**. The updated data frame may be forwarded to network component **140** via second edge interface circuit **149** at egress port **144**.

(17) Network component **130** may be coupled to edge circuit **135**. Edge circuit **135** may include a first edge interface circuit **138** to communicate with network component **130** using a store-and-forward protocol. First edge interface circuit **138** may communicate with network component **130** via ingress port **132** and egress port **131**. Edge circuit **135** may include one or more internal interface circuits **136**, **137** to communicate with internal network segment **150**.

(18) Edge circuit **135** may receive a frame of data at first edge interface circuit **138** via ingress port **132**. Edge circuit **135** may include an error detector circuit **192** to detect errors in the received data frame. Error detector circuit **192** may perform a CRC check or other error detection check on the received data frame. The frame of data received at edge circuit **135** may contain identification information. Edge circuit **135** may prepend a frame-specific information header in the received data frame to create a prepended data frame. The frame-specific information header may be generated by an identification circuit **195**. Identification circuit **195** may be implemented in hardware or software, or a combination thereof. The frame-specific information header may include an identifier associated with the received data frame. This identifier may also be termed a frame number. The frame-specific information header may be part of a tag prepended to the received data frame to generate a prepended data frame. The tag may be prepended by tag insertion circuit **196**.

Tag insertion circuit **196** may be coupled to identification circuit **195**, and may be implemented in hardware or software, or a combination thereof. Tag insertion circuit **196** may be integrated into first edge interface circuit **138**, or may be a separate component. The prepended data frame with the prepended frame-specific information header may be forwarded to edge circuit **145** via forwarder circuit **133**, internal interface circuit **137** and path **164**. The prepended data frame with the prepended frame-specific information header may be forwarded to cut-through forwarder circuit **151** of internal network segment **150** via forwarder circuit **133**, internal interface circuit **137** and path **165**. The prepended data frame with the prepended frame-specific information header may be forwarded to cut-through forwarder circuit **152** of internal network segment **150** via forwarder circuit **133**, internal interface circuit **136** and path **163**.

(19) Edge circuit **135** may receive a frame of data at first internal interface circuit **136** via path **163**. Edge circuit **135** may include an error detector circuit **192** to detect errors in the received data frame and discard data frames with a detected error. Error detector circuit **192** may perform a CRC check or other error detection check on the received data frame. If an error is detected, the frame may be discarded. The frame of data received at edge circuit **135** may contain identification information. Edge circuit **135** may update portions of the received data frame and forward the updated data frame to network component **130** using a S&F protocol via egress port **131**. Edge circuit **135** may read the portion of the received data frame comprising the frame-specific information header. The frame-specific information header may be present at a predetermined location in the received data frame. Edge circuit **135** may perform a lookup operation to determine if the frame-specific information header matches the frame-specific information header of a previous data frame that has already been forwarded. The frame-specific information header may be stored in RDT **191** in edge circuit **135**. RDT **191** may be a non-transitory memory. If the frame-specific information header of the received data frame is listed in RDT **191**, the received data frame may be discarded and not forwarded since it has already been forwarded through egress port **131**. The lookup operation and discarding of duplicate data frames may be implemented by a duplicates removal circuit **193** of edge circuit **135**. The duplicates removal circuit may be implemented in hardware or software, or a combination thereof. If the frame-specific information header of the received data frame is not listed in RDT **191**, the data frame may be termed a non-duplicate data frame, and edge circuit **135** may add the frame-specific information header to RDT **191**. After performing the lookup operation, edge circuit **135** may remove the frame-specific information header from the data frame to create an updated data frame. The removal of the frame-specific information header may be implemented by a tag removal circuit **194** of edge circuit **135**. The tag removal circuit **194** may be implemented in hardware or software, or a combination thereof. The updated data frame may be forwarded to edge interface circuit **138** via forwarder circuit **133**. The updated data frame may be forwarded to network component **130** via edge interface circuit **138** at egress port **131**. Error detector circuit **192** may be included in duplicates removal circuit **193**, or may be a separate circuit.

(20) Edge circuit **135** may receive a frame of data at internal interface circuit **137** via path **164**. Edge circuit **135** may receive data at internal interface circuit **137** via path **165**. Edge circuit **135** may include an error detector circuit **192** to detect errors in the received data frame and discard data frames with a detected error. Error detector circuit **192** may perform a CRC check or other error detection check on the received data frame. If an error is detected, the frame may be discarded. The frame of data received at edge circuit **135** may contain identification information. Edge circuit **135** may update portions of the received data frame and forward the updated data frame to network component **130** using a S&F protocol via egress port **131**. Edge circuit **135** may read the portion of the received data frame comprising the frame-specific information header. Edge circuit **135** may perform a lookup operation in cooperation with RDT **191** to determine if the frame-specific information header matches the frame-specific information header of a previous data frame that has already been forwarded. If the frame-specific information header of the

received data frame is listed in RDT **191**, the received data frame may be discarded and not forwarded since it has already been forwarded through egress port **131**. If the frame-specific information header of the received data frame is not listed in RDT **191**, the data frame may be termed a non-duplicate data frame, and edge circuit **135** may add the frame-specific information header to RDT **191**. After performing the lookup operation, edge circuit **135** may remove the frame-specific information header from the data frame to create an updated data frame. The removal of the frame-specific information header may be implemented by tag removal circuit **194** in edge circuit **135**. The updated data frame may be forwarded to edge interface circuit **138** via forwarder circuit **133**. The updated data frame may be forwarded to network component **130** via edge interface circuit **138** via egress port **131**.

(21) The example illustrated in FIG. **1** is merely for illustrative purposes and is not intended to limit the invention. The example illustrated in FIG. **1** includes a specific number of network components, edge interface circuits, internal interface circuits, and network components, cut-through forwarder circuits of internal network segments and of edge circuits. The specific paths between network components, edge interface circuits, internal interface circuits, cut-through forwarder circuits, and edge circuits illustrated in FIG. **1** is for illustrative purposes and is not intended to limit the invention.

(22) Other examples not illustrated in FIG. **1** may include more network components than the number of network components illustrated in FIG. **1**. Other examples not illustrated in FIG. **1** may include fewer network components than the number of network components illustrated in FIG. **1**.

(23) Other examples not illustrated in FIG. **1** may include more edge circuits than the number of edge circuits illustrated in FIG. **1**. Other examples not illustrated in FIG. **1** may include fewer edge circuits than the number of edge circuits illustrated in FIG. **1**.

(24) Other examples not illustrated in FIG. **1** may include more edge interface circuits than the number of edge interface circuits illustrated in FIG. **1**. Other examples not illustrated in FIG. **1** may include fewer edge interface circuits than the number of edge interface circuits illustrated in FIG. **1**.

(25) Other examples not illustrated in FIG. **1** may include more internal interface circuits than the number of internal interface circuits illustrated in FIG. **1**. Other examples not illustrated in FIG. **1** may include fewer internal interface circuits than the number of internal interface circuits illustrated in FIG. **1**.

(26) Other examples not illustrated in FIG. **1** may include more network components than the number of network components illustrated in FIG. **1**. Other examples not illustrated in FIG. **1** may include fewer network components than the number of network components illustrated in FIG. **1**.

(27) Other examples not illustrated in FIG. **1** may include different network components than the specific network components illustrated in FIG. **1**.

(28) FIG. **2** illustrates one of various examples of an edge circuit **200**. Edge circuit **200** may represent one of various examples of edge circuit **145** as illustrated in FIG. **1**. In the example of FIG. **2**, first edge interface circuit **270** and second edge interface circuit **275** may interface with a network component **201**, and first internal interface circuit **280** and second internal interface circuit **285** may interface with internal network segment **202**. Internal network segment **202** may be one of various examples of internal network segment **150**, as illustrated and described in reference to FIG. **1**.

(29) Edge circuit **200** may include RDT **290**. RDT **290** may be a non-transitory memory. In one of various examples, first edge interface circuit **270**, second edge interface circuit **275**, first internal interface circuit **280** and second internal interface circuit **285** may respectively include an RDT. In one of various examples, an RDT may be shared by more than one of first edge interface circuit **270**, second edge interface circuit **275**, first internal interface circuit **280** and second internal interface circuit **285**.

(30) First edge interface circuit **270** may include error detector circuit **296**. Error detector circuit **296** may generate an output signal based on detection of an error. Second edge interface circuit **275**

may include error detector circuit **297**. Error detector circuit **297** may generate an output signal based on detection of an error. First internal interface circuit **280** may include error detector circuit **298**. Error detector circuit **298** may generate an output signal based on detection of an error. Second internal interface circuit **285** may include error detector circuit **299**. The example illustrated in FIG. 2 illustrates a separate error detector circuit for, respectively, first edge interface circuit **270**, second edge interface circuit **275**, first internal interface circuit **280** and second internal interface circuit **285**, but this is not intended to be limiting. In one of various examples, a single error detector circuit may be shared by more than one of first edge interface circuit **270**, second edge interface circuit **275**, first internal interface circuit **280** and second internal interface circuit **285**.

(31) Forwarder circuit **205** may receive data from first edge interface circuit **270**, second edge interface circuit **275**, first internal interface circuit **280** and second internal interface circuit **285** and may forward the received data to the proper destination.

(32) First edge interface circuit **270** may receive a data frame from forwarder circuit **205** via path **210**. The data frame may be received by error detector circuit **296**. Error detector circuit **296** may check for errors in the data frame and may discard frames with detected errors. Checking for errors may be a CRC check or another error check technique. Error detector circuit **296** may detect errors in the data frame and may provide an error output to duplicates removal circuit **211**. Duplicates removal circuit **211** may remove data frames with errors based on the error output. Duplicates removal circuit **211** may include a controller circuit to perform a lookup operation to determine if the frame-specific information header of the received frame of data matches the frame-specific information header of a previous frame of data that has already been forwarded at egress port **272**. Alternately, operation of duplicates removal circuitry **211** may be responsive to controller **295**. If the frame-specific information header of the received data is listed in RDT **290**, the received frame is discarded by duplicates removal circuit **211** and not forwarded since it has already been forwarded through egress port **272**. If the frame-specific information header of the received data is not listed in RDT **290**, the data frame may be termed a non-duplicate data frame, and controller circuit **295** may add the frame-specific information header to RDT **290**. Tag removal circuit **212** may remove the frame-specific information header from the frame and forward the updated frame from first edge interface circuit **270** to network component **201** at egress port **272**. Error detector circuit **296** may be included in duplicates removal circuit **211**, or may be a separate circuit as illustrated in FIG. 2.

(33) First edge interface circuit **270** may receive a frame of data from network component **201** via ingress port **271**. Ingress port **271** may be coupled to identification circuit **222**. Identification circuit **222** may prepend a frame-specific information header to the received data frame. The frame-specific information header may include an identifier associated with the received data frame. This identifier may also be termed a frame number. The identifier or frame number may distinguish a first received data frame, received from a particular MAC address, from a second received data frame received from the same MAC address at an earlier or later point in time. Identification circuit **222** may be coupled to tag insertion circuit **221**. Tag insertion circuit **221** may combine the frame-specific information header with the received data frame to create a prepended data frame. The prepended data frame may be forwarded through path **220** to forwarder circuit **205** for forwarding to other destinations on the network.

(34) Second edge interface circuit **275** may receive a frame of data from forwarder circuit **205** via path **230**. Network component **201** may interface to second edge interface circuit **275**. The data frame may be received by error detector circuit **297**. Error detector circuit **297** may check for errors in the data frame and may provide an error output to duplicates removal circuit **231**. Checking for errors may be a CRC check or another error check technique. Duplicates removal circuit **231** may remove data frames with errors based on the error output. Duplicates removal circuit **231** may include a controller circuit to perform a lookup to determine if the frame-specific information header of the received frame of data matches the frame-specific information header of a previous

frame of data that has already been received. Alternately, operation of duplicates removal circuit **231** may be responsive to controller **295**. If the frame-specific information header of the received data is listed in RDT **290**, the received frame is discarded by duplicates removal circuit **231** and not forwarded since it has already been forwarded through egress port **277**. If the frame-specific information header of the received data is not listed in RDT **290**, the data frame may be termed a non-duplicate data frame, and controller circuit **295** may add the frame-specific information header to RDT **290**. Tag removal circuit **232** may remove the frame-specific information header from the frame and forward the updated frame from second edge interface circuit **275** at egress port **277**. Error detector circuit **297** may be included in duplicates removal circuit **231**, or may be a separate circuit as illustrated in FIG. 2.

(35) Second edge interface circuit **275** may receive a frame of data via ingress port **276**. Ingress port **276** may be coupled to identification circuit **242**. Identification circuit **242** may prepend a frame-specific information header to the received frame. The frame-specific information header may include an identifier associated with the received data frame. This identifier may also be termed a frame number. The identifier or frame number may distinguish a first received data frame, received from a particular MAC address, from a second received data frame received from the same MAC address at an earlier or later point in time. Identification circuit **242** may be coupled to tag insertion circuit **241**. Tag insertion circuit **241** may prepend the frame-specific information header to the received data to create a prepended data frame. The prepended data frame including the frame-specific information header may be forwarded through path **240** to forwarder circuit **205** for forwarding to other destinations on the network.

(36) Edge circuit **200** may receive a data frame from internal network segment **202** at first internal interface circuit **280** via first ingress port **282**. The received data frame may be forwarded to forwarder circuit **205** via first egress port **250**.

(37) First internal interface circuit **280** may forward a data frame over second egress port **281**. The data frame may be received from forwarder circuit **205** via second ingress port **255**. The data frame may be received by error detector circuit **298**. Error detector circuit **298** may check for errors in the data frame and may provide an error output to duplicates removal circuit **251**. Checking for errors may be a CRC check or another error check technique. Duplicates removal circuit **251** may remove data frames with errors based on the error output. Duplicates removal circuit **251** may read the portion of the received data frame comprising the frame-specific information header. Duplicates removal circuit **251** may include a controller circuit to perform a lookup operation and determine if the frame-specific information header of the received frame of data matches the frame-specific information header of a previous frame of data that has already been received and forwarded by second egress port **281**. Alternately, operation of duplicates removal circuit **251** may be responsive to controller **295**. If the frame-specific information header of the received data is listed in RDT **290**, the received frame is discarded and not forwarded since it has already been forwarded through second egress port **281**. If the frame-specific information header of the received data is not listed in the RDT **290**, the data frame may be termed a non-duplicate data frame, and controller circuit **295** may add the information to the RDT **290**. First internal interface circuit **280** may forward the frame via second egress port **281**. Error detector circuit **298** may be included in duplicates removal circuit **251**, or may be a separate circuit as illustrated in FIG. 2.

(38) Edge circuit **200** may receive a data frame from internal network segment **202** at second internal interface circuit **285** via first ingress port **287**. Received data may be forwarded to forwarder circuit **205** via first egress port **260**.

(39) Second internal interface circuit **285** may forward a data frame over second egress port **286**. Data may be received from forwarder circuit **205** via second ingress port **265**. The data frame may be received by error detector circuit **299**. Error detector circuit **299** may check for errors in the data frame and may provide an error output to duplicates removal circuit **261**. Checking for errors may be a CRC check or another error check technique. Duplicates removal circuit **261** may remove data

frames with errors based on the error output. Duplicates removal circuit **261** may read the portion of the received data frame comprising the frame-specific information header. Duplicates removal circuit **261** may include a controller circuit to perform a lookup to determine if the frame-specific information header matches the frame-specific information of a previous frame of data that has already been forwarded by second egress port **286**. Alternately, operation of duplicates removal circuit **231** may be responsive to controller **295**. If the frame-specific information header of the received data is listed in RDT **290**, the received frame is discarded and not forwarded since it has already been forwarded through second egress port **286**. If the frame-specific information header of the received data is not listed in the RDT **290**, the data frame may be termed a non-duplicate data frame, and controller circuit **295** may add the frame-specific information header to RDT **290**. Second egress port **286** may forward the data to internal network segment **202**. Error detector circuit **299** may be included in duplicates removal circuit **261**, or may be a separate circuit as illustrated in FIG. 2.

(40) FIG. 3 illustrates a method for control of network traffic. A device as previously illustrated and described in reference to FIG. 2 may execute the illustrated method.

(41) At operation **305**, a data frame may be received by an edge interface circuit from a network component.

(42) At operation **310**, the edge interface circuit may generate frame identification information associated with the data frame. The frame-specific information header may include an identifier for the received data frame. This identifier may also be termed a frame number. The identifier or frame number may distinguish a first received data frame, received from a particular MAC address, from a second received data frame received from the same MAC address at an earlier or later point in time.

(43) At operation **315**, frame-specific information header may be prepended as a tag onto the received data frame to generate a prepended data frame.

(44) At operation **320**, the prepended data frame may be forwarded to a forwarder circuit.

(45) At operation **325**, a data frame may be received at an edge interface circuit from the forwarder circuit. The data frame received at the edge interface circuit may be a prepended data frame as described in reference to operations **305** through **320**, inclusive. An error detection operation may be performed on the received data frame and data frames with detected errors may be removed.

(46) At operation **330**, duplicate data frames received at the edge interface circuit may be selectively removed. The frame-specific information header may be read from a portion of the received data frame. The frame-specific information header may be present at a predetermined location in the received data frame. A lookup operation may determine if the frame-specific information header in the received data frame matches the frame-specific information header of a previous data frame that has already been forwarded. The frame-specific information header may be stored in an RDT. If the frame-specific information header of the received data frame is listed in the RDT, the received data frame may be discarded and not forwarded since it has already been forwarded.

(47) At operation **335**, the frame-specific information header may be removed from the received data frame to create an updated data frame if the received data frame is not a duplicate.

(48) At operation **340**, the updated data frame may be forwarded to a network component through an egress port.

(49) At operation **345**, a data frame may be received at an internal interface circuit from a network component. At operation **350**, the received data frame may be forwarded to a forwarder circuit.

(50) At operation **355**, a data frame may be received at an internal interface circuit from a forwarder circuit. The data frame received at the internal interface circuit may be a prepended data frame as described in reference to operations **305** to **320**, inclusive. An error detection operation may be performed on the received data frame and data frames with detected errors may be removed. At operation **360**, if the received data frame is a duplicate data frame, the received data

frame may be selectively removed. Frame-specific information header may be read from a portion of the received data frame. Frame-specific information header may be present at a predetermined location in the received data frame. A lookup operation may determine if the frame-specific information header in the received data frame matches the frame-specific information header of a previous data frame that has already been forwarded. The frame-specific information header may be stored in an RDT. If the frame-specific information header of the received data frame is listed in the RDT, the received data frame may be discarded and not forwarded since it has already been forwarded.

(51) At operation **365**, non-duplicate received data frames may be forwarded to a network component.

Claims

1. A device comprising: an edge interface circuit to couple to a network component and to couple to a forwarder circuit, the edge interface circuit to: receive data frames from the network component, prepend a frame-specific information header to the received data frames and forward the prepended data frames to the forwarder circuit; receive data frames from the forwarder circuit, remove duplicate data frames and forward non-duplicate data frames to the network component; an internal interface circuit to couple to an internal network segment and to couple to the forwarder circuit, the internal interface circuit to: receive data frames from the internal network segment and forward the received data frames to the forwarder circuit, and receive data frames from the forwarder circuit, remove duplicate data frames and forward non-duplicate data frames to the internal network segment.
2. The device as claimed in claim 1, the edge interface circuit comprising: an ingress port to couple to the network component, the ingress port to receive data frames; an identification circuit to couple to the first ingress port, the identification circuit to generate a frame-specific information header based at least on the received data frames, and a tag insertion circuit to couple to the identification circuit, the tag insertion circuit to prepend the frame-specific information header to the received data frames to generate prepended data frames, the tag insertion circuit to forward the prepended data frames to the forwarder circuit.
3. The device as claimed in claim 2, the edge interface circuit comprising: an error detector circuit to receive data frames from the forwarder circuit; a duplicates removal to remove duplicate data frames based at least on the contents of a non-transitory memory and the frame-specific information header; a tag removal circuit to couple to the duplicates removal circuit, the tag removal circuit to remove the frame-specific information header in received data frames to generate updated data frames, and an egress port to couple to the tag removal circuit, the egress port to forward the updated data frames to the network component.
4. The device as claimed in claim 3, wherein the duplicates removal circuit comprises a controller circuit to search the contents of the non-transitory memory for the presence of a portion of a received data frame within the non-transitory memory contents.
5. The device as claimed in claim 3, the duplicates removal circuit to remove received data frames based on an output of the error detector circuit.
6. The device as claimed in claim 1, the internal interface circuit comprising: a first ingress port to couple to an internal network segment, and to couple to the forwarder circuit, the first ingress port to receive data frames from the internal network segment and to forward received data frames to the forwarder circuit through a first egress port; a second ingress port to couple to the forwarder circuit, the second ingress port to receive data frames from the forwarder circuit; an error detector circuit to couple to the second ingress port; a duplicates removal circuit to couple to the error detector circuit, the duplicates removal circuit to remove duplicate received data frames based at least on the contents of a non-transitory memory and the frame-specific information header, and a

second egress port to forward non-duplicate data frames to the internal network segment.

7. The device as claimed in claim 6, wherein the duplicates removal circuit comprises a controller circuit to search the contents of the non-transitory memory for the presence of a portion of the received data frame within the non-transitory memory contents.

8. The device as claimed in claim 6, the duplicates removal circuit to remove received data frames based on an output of the error detector circuit.

9. A system comprising: a network comprising: a plurality of network connected devices, comprising: an edge interface circuit to couple to a network component and to couple to a forwarder circuit, the edge interface circuit to: receive data frames from the network component, prepend a frame-specific information header to the received data frames and forward the prepended data frames to the forwarder circuit; receive data frames from the forwarder circuit, remove duplicate data frames and forward non-duplicate data frames to the network component; an internal interface circuit to couple to an internal network segment and to couple to the forwarder circuit, the internal interface circuit to: receive data frames from the internal network segment and forward the received data frames to the forwarder circuit, and receive data frames from the forwarder circuit, remove duplicate data frames and forward non-duplicate data frames to the internal network segment.

10. The system as claimed in claim 9, the edge interface circuit comprising: an ingress port to couple to the network component, the ingress port to receive data frames; an identification circuit to couple to the first ingress port, the identification circuit to generate a frame-specific information header based at least on the received data frames, and a tag insertion circuit to couple to the identification circuit, the tag insertion circuit to prepend the frame-specific information header to the received data frames to generate prepended data frames, the tag insertion circuit to forward the prepended data frames to the forwarder circuit.

11. The system as claimed in claim 10, the edge interface circuit comprising: an error detector circuit to receive data frames from the forwarder circuit; a duplicates removal circuit to receive data frames from the error detector circuit, the duplicates removal circuit to remove duplicate data frames based at least on the contents of a non-transitory memory and the frame-specific information header; a tag removal circuit to couple to the duplicates removal circuit, the tag removal circuit to remove the frame-specific information header in received data frames to generate updated data frames, and an egress port to couple to the tag removal circuit, the egress port to forward the updated data frames to the network component.

12. The system as claimed in claim 11, wherein the duplicates removal circuit comprises a controller circuit to search the contents of the non-transitory memory for the frame-specific information header within the non-transitory memory contents.

13. The system as claimed in claim 11, the duplicates removal circuit to remove received data frames based on an output of the error detector circuit.

14. The system as claimed in claim 11, the internal interface circuit comprising: a first ingress port to couple to an internal network segment, and to couple to the forwarder circuit, the first ingress port to receive data frames from the internal network segment and to forward the received data frames to the forwarder circuit through a first egress port; a second ingress port to couple to the forwarder circuit, the second ingress port to receive data frames from the forwarder circuit; an error detector circuit to couple to the second ingress port; a duplicates removal circuit to couple to the error detector circuit, the duplicates removal circuit to remove duplicate received data frames based at least on the contents of a non-transitory memory and the frame-specific information header, and a second egress port to forward non-duplicate data frames to the internal network segment.

15. The system as claimed in claim 14, wherein the duplicates removal circuit comprises a controller circuit to search the contents of the non-transitory memory for the presence of the frame-specific information header within the non-transitory memory contents.

16. The system as claimed in claim 14, the duplicates removal circuit to remove received data

frames based on an output of the error detector circuit.

17. A method comprising: receiving a data frame at an edge interface circuit from a network component; generating a frame-specific information header based on the received data frame; prepending the frame-specific information header as a tag onto the received data frame to generate a prepended data frame; forwarding the prepended data frame to a forwarder circuit; receiving a data frame at an edge interface circuit from the forwarder circuit and discarding data frames with a detected error; identifying, based upon contents of a non-transitory memory and the frame-specific information header, duplicate received data frames; removing duplicate received data frames; removing the frame-specific information header from the non-duplicate received data frame to create an updated data frame; forwarding the updated data frame to a network destination through an egress port; receiving a data frame at an internal interface circuit from an internal network segment; forwarding the received data frame to the forwarder circuit; receiving a data frame at an internal interface circuit from the forwarder circuit; identifying, based upon contents of a non-transitory memory and the frame-specific information header, duplicate received data frames; removing duplicate received data frames, and forwarding non-duplicate data frames to a network component.

18. The method as claimed in claim 17, wherein removing duplicate received data frames comprises a controller circuit to search the contents of the non-transitory memory for the frame-specific information header within the non-transitory memory contents.
